

KaraokeVUI: Utilizing *Karaoke* Subtitles for Voice User Interfaces to Navigate Users What They Would Say

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ABSTRACT

Voice user interfaces (VUIs) have the problems in discoverability and learnability. Conventional VUIs tried to solve these problems by listing voice commands for using features on the screen of devices in combination with GUIs. However, with these VUI help tools, it is difficult to know how accurately the commands must be spoken and to determine whether the target operation is being executed if the words are rephrased or spoken incorrectly during speaking. Therefore, users are too preoccupied with speaking phrases correctly, and the advantage of VUIs, which can be used without awareness of the operation method, has not been fully demonstrated. To address these issues, we propose KaraokeVUI, which is a VUI help tool for supporting voice operation with feedback by displaying spoken words through filling in the blanks or overlaying on phrases, just like on a karaoke screen. In this paper, we evaluate and verify KaraokeVUI usefulness and usability.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction, Interaction design.;

KEYWORDS

Voice user interfaces, user experience, accessibility

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1 INTRODUCTION

Many of the applications for smartphones provided by our company are equipped with a voice search software development kit (SDK) developed in-house for Japanese speakers. In the current SDK, the main feature is voice search by keywords. There are two types of search tasks: known-item search and exploratory search [1], and keyword search is not suitable for exploratory search, in which users do not fully understand the query [2]. Exploratory search often involves searching for ambiguous keywords then narrowing the search target [1], and voice search is also likely to be used in this

way in the future. From the above, we are considering updating the SDK so that advanced search features, such as narrowing, changing search parameters, and sorting search results, can be used by voice.

To make the aforementioned advanced features available by voice operation, we have to face the discoverability and learnability problem [4], which current voice user interfaces (VUIs) have. Discoverability is the ability for users to find and use features through a UI, and learnability means to the ability of novice users to easily learn how to use a new system without prior training [7, 8]. When using VUIs, it is more difficult to form mental models that appropriately map features and operations than using graphical UIs (GUIs) [3], so it is difficult for users to imagine what features are available and learn what to say to operate them [8]. Therefore, we propose KaraokeVUI, which is a VUI help tool for supporting voice operation that is effective in addressing these issues and can be used with low cognitive loads.

2 RELATED WORK

To approach the discoverability and learnability problem with VUIs, multimodal interfaces combined with GUIs, which are interfaces with high perspicuity, have been considered effective [9]. In Voice-Navigator [8] and JustSpeak [10], numbers or feature names are labeled near the icons to indicate which features are available by voice operation and to unify the names for voice operation. Voice-Navigator [8] and DiscoverCal's "What can I do?" menu [4] use a technique that displays a list of voice commands that can be operated by voice when the user says "Help" or "What can I say?". In particular, DiscoverCal uses the UI that displays a list of voice commands according to the user's proficiency level, and recent research has adopted a method of display that also predicts the user's goal [6]. In the usability verification of DiscoverCal, errors during operation and the user's tactics to deal with them were examined in detail [5], but it has not been examined the cognitive load of using the VUIs while using these help tools. Therefore, we re-implemented a voice-search application for smartphones using current VUI help tools [4] and conducted a face-to-face survey and user interview. According to this survey, while using these help tools, it seems that users must be aware of speaking accuracy, which placed extra cognitive loads on them because of the difficulty in grasping how much error can be allowed for blurring of speech content, and how correctly their speech is recognized for the target operation. We considered that the above problems could be solved by providing appropriate feedback during speech when displaying the help menu. KaraokeVUI is a feedback-VUI help tool that is effective in reducing such cognitive loads in a help-menu screen and supports the discoverability and learnability of VUIs.

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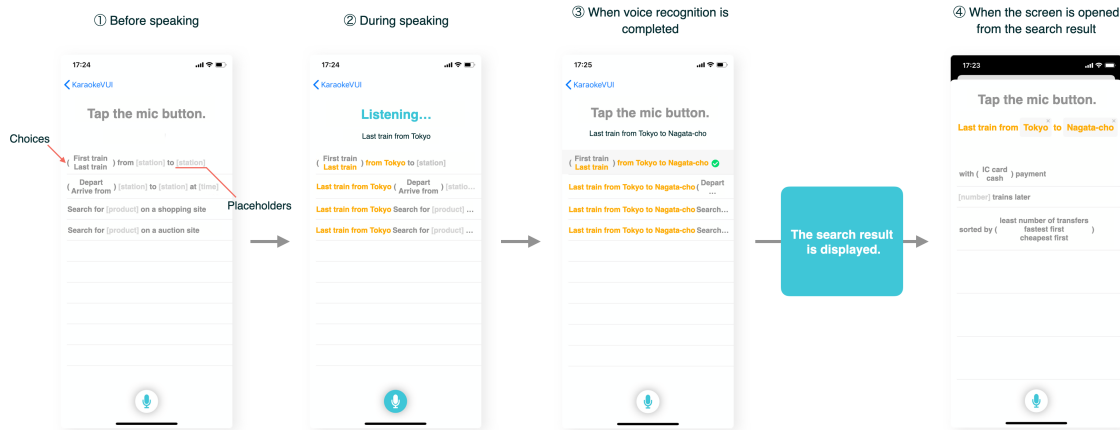


Figure 1: Screen when using KaraokeVUI built for English grammar

3 KARAOKE VUI

3.1 Interaction Design

KaraokeVUI displays feedback on which operation the current utterance corresponds to when users use a help tool that supports improving discoverability and learnability of VUIs. KaraokeVUI is a VUI help tool in which spoken words fill in blanks or overlay phrases that is provided from the help tool, just like a karaoke screen view (Figure 1).

Before a user speaks, this help tool displays a list of features available for voice control and effective speech examples to use them (Figure 1①). When he starts speaking in this state, as shown in Figure 1②, a spoken sentence is displayed as a fill-in-the-blank or overlay in orange text. When he takes a breath and finishes speaking, the feature has been actually requested will be highlighted and a check mark will appear at the end of the sentence (Figure 1③). After this view is displayed, the checked feature is executed (in this case, the transit search is executed and the search result of "Last train from Tokyo to Nagata-cho" is displayed). Since a prototype of KaraokeVUI was implemented for exploratory search, we also implemented a feature to inherit the context of the previous search. When using voice search from the search-result screen again, the previous search content is inherited, and the user can change queries and narrow the search results (Figure 1⑤).

The prototype used in this experiment begins with the screen showing a list of speech examples for the convenience of the experiment, but the interaction in the actual application is supposed to be displayed in situations where the user is not familiar with voice operation, such as when he does not start speaking for a while or taps the help button.

3.2 System Design

The application side holds three types of data: placeholders, particles, and keywords that are used as criteria for clipping placeholders, to display KaraokeVUI. To increase the visibility of partially the same phrases, we devised a bifurcated notation (Figure 1). The icon of a check mark is displayed on the example phrase with the highest degree of conformity after the user has finished speaking and all

placeholders have been filled in for the parameters required for the operation.

For the programming logic to display text for KaraokeVUI, it was necessary to build it for each language since word orders and grammar differ among them. In this experiment, we built it for Japanese grammar since we were conducting the experiment with Japanese speakers.

4 USER EVALUATIONS

The participants were 96 Japanese speakers in their teens to 60s recruited through crowdsourcing. We distributed materials and applications for this experiment and had the participants participate individually in a remote environment. It was a within-subjects designed experiment to compare the usability of a current VUI help tool and KaraokeVUI for the same user. After checking the operation and practicing 4 test tasks, the participants were asked to perform the production tasks as novice users. We regard the VUI help tool imitated DiscoverCal's "What can I do?" menu [4] as the baseline, and designed the respective screens of the baseline and KaraokeVUI with the specifications shown in Figure 2. We distributed them as separate smartphone applications and conducted experiments for user evaluation.

There were 13 production tasks in each of the patterns, and the participants were asked to perform voice operations as required in the task. We conducted both quantitative evaluation by analyzing the quantitative data of users' operations and qualitative evaluation by analyzing the results of questionnaires that were answered after operations. To avoid differences in operation caused by factors other than differences in the UI, the same task was performed for each pattern. However, since this experiment was a within-subjects experiment, there was a possibility that the tasks of the pattern performed latter would produce more favorable results due to habituation of operation. Therefore, we divided the participants into two groups, one starting with the baseline and the other starting with KaraokeVUI.

4.1 Quantitative Evaluation

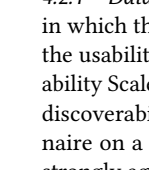
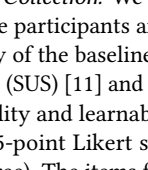
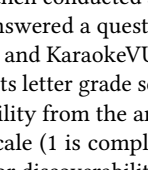
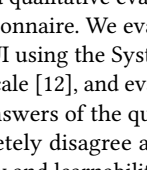
the baseline				KaraokeVUI			
- The VUI help tool displays a list of available voice commands. - From the second time of search, the search parameters are inherited, and it is possible to change parameters and narrow the search.				- The VUI help tool in which spoken words fill in blanks or overlay phrases, just like a karaoke screen view. - Display using bifurcated notation and placeholders. - From the second time of search, the search parameters are inherited, and it is possible to change parameters and narrow the search.			
① Before speaking	② During speaking	③ When voice recognition is completed	④ Second time of search	① Before speaking	② During speaking	③ When voice recognition is completed	④ Second time of search
							

Figure 2: Patterns and conditions of this experiment

4.1.1 Data Collection. We conducted this experiment with the baseline and KaraokeVUI, as shown in Figure 2, and collected a total of 2496 sets of data (13 tasks*2 types of applications * 96 people). The following four types of data were collected as one set for this evaluation.

- (a) **The number of task failures:** the number of times the participant failed a task due to a mistake in speech, operation, etc.
- (b) **The waiting time from opening the voice-operation screen until speech is started:** the time between being shown the list of speech examples and starting to speak (the time the user is thinking about how to speak).
- (c) **The waiting time from tapping the microphone button until speech is started:** the time between the participant deciding the content of speech and starting to speak.
- (d) **The time spent on speech:** the time between tapping the microphone button and end of voice input.

4.1.2 Results. We conducted a paired t-test to investigate whether KaraokeVUI reduce cognitive loads more than the baseline. The results in Table 1 indicate that the average number of task failures is a positive number when this number for KaraokeVUI is subtracted from that for the baseline. Since this result is significantly different even at the 1% level of significance, we can see that the number of voice-operation failures can be reduced with KaraokeVUI ($t(1247) = 2.93, p < 0.01$, Cohen's $d = 0.0831$).

Regarding "(b) The waiting time from opening the voice-operation screen until speech is started", KaraokeVUI reduced the waiting time ($t(1247) = 4.53, p < 0.01$, Cohen's $d = 0.128$). However, there was no significant difference in "(c) The waiting time from tapping the microphone button until speech is started." ($t(1247) = 1.07$, n.s.) We analyzed the videos of participants' operations and found that they performed voice operation in the following flow: open the help screen, look at the list and think about what to say, tap the microphone button, and speak. Of this flow, there was no significant difference between the current VUI help tool and KaraokeVUI in the time from tapping the microphone button until starting to speak, on the other hand, there was a difference in the time from opening the voice-operation screen until starting to speak. In other words, we can say that the difference occurs in the time, which is the difference between these two times, while thinking about what to say. Therefore, KaraokeVUI reduces the

time to think about what to say for voice operation and reduces cognitive load while thinking about it.

4.2 Qualitative Evaluation

4.2.1 Data Collection. We then conducted a qualitative evaluation in which the participants answered a questionnaire. We evaluated the usability of the baseline and KaraokeVUI using the System Usability Scale (SUS) [11] and its letter grade scale [12], and evaluated discoverability and learnability from the answers of the questionnaire on a 5-point Likert scale (1 is completely disagree and 5 is strongly agree). The items for discoverability and learnability were as follows.

The Questions items are:

1. With KaraokeVUI, it is easier to find the speech utterances that correspond to the search operation you want to use than with the baseline.
2. With KaraokeVUI, it is easier to understand whether you could speak correctly in response to the examples of speech than with the baseline.
3. There is less anxiety during speech and operation with KaraokeVUI than with the baseline.
4. KaraokeVUI is more visually clear and less confusing than the baseline.
5. KaraokeVUI is overall more useful than the baseline.

4.2.2 Results. In terms of SUS scores, the baseline was given a grade of D (Poor) [12] with an average of 63.4 points, and KaraokeVUI was given a B (Good) [12] with an average of 68.8 points. These results indicate that the current VUI help tool has low usability, and KaraokeVUI has higher than average usability. We also analyzed whether discoverability and learnability were improved in terms of reducing cognitive loads through a questionnaire on a Likert scale, and as shown in Table 2, all items were above 3.0, indicating that KaraokeVUI improves these more than the current VUI help tool.

5 DISCUSSION

If the feedback is given during speech, users may be influenced by that, and may change the content of speech while speaking. This is like the "synlogue" [13] style of speaking, which is considered to be a characteristic way of speaking in Japanese conversation. "Synlogue" is a way of speaking that can be considered in contrast to "dialogue." If dialogue is a way of speaking in which speakers speak independently and take turns, synlogue is a way of speaking in which speakers build up a conversation together while overlapping and taking over each other's speech processes. Thus, in the Japanese language, a conversation is thrown to the other person in an unfinished state, and the conversation is built together on the basis of the other person's response, such as gestures and expressions, so that if there is no response during the speech, it tends to be difficult to speak. Feedback such as with KaraokeVUI, which shows the prediction of the speech content during speaking, may be a more natural interaction for Japanese speakers. Interesting results should be obtained by comparing and verifying with other languages.

Table 1: Summary of quantitative evaluation results

		Average(the baseline)	Average(KaraokeVUI)	t value	p value	Effects size(Cohen's d)
	(a)	0.3569	0.2269	2.93	p<.01	0.0831
(b)	Include failed tasks	4.197	3.520	4.53	p<.01	0.128
	Only successful tasks	4.482	3.690	4.04	p<.01	0.114
(c)	Include failed tasks	0.003438	0.003173	1.07	n.s.	0.0304
	Only successful tasks	0.003419	0.003138	1.14	n.s.	0.0323
(d)	Include failed tasks	4.121	4.023	3.9	p<.01	0.110
	Only successful tasks	4.068	3.990	3.18	p<.01	0.0903

Table 2: Result of discoverability and learnability questionnaire

	Q1	Q2	Q3	Q4	Q5
Average score	3.34	3.55	3.46	3.41	3.41

6 CONCLUSION

We proposed KaraokeVUI, which is a VUI help tool that provides feedback to indicate tolerance of speech inaccuracies and degree of conformity while users are speaking and reported the results of a quantitative and qualitative usability evaluations. We found that KaraokeVUI is effective in improving the learnability and usability of VUIs for Japanese speakers. Since the results may be different for other languages, further verification is necessary.

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